

Accessible LaTeX example

Overleaf Support

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Abstract

A simple $\text{\LaTeX}$ document authored using Overleaf. This document takes advantage of the PDF tagging support provided by the $\text{\LaTeX}$ team that is available in TeXLive 2025.

1 Introduction

This simple example describes some of the considerations that go into creating accessible documents with $\text{\LaTeX}$ using Overleaf.

1.1 Selecting the TeX Live version in Overleaf

Overleaf uses a standard TeX Live $\text{\LaTeX}$ environment, making the accessibility support that is being provided by the $\text{\LaTeX}$ team available to all Overleaf users.

With 2025 TeX Live, $\text{\LaTeX}$ can now produce tagged PDFs automatically – a key requirement for accessibility. Additional accessibility features are being added on an ongoing basis to the TeX Live builds. Currently, new projects created in Overleaf use the 2025 TeX Live tested release, and for those who need even more recent releases, a rolling updated version of TeX Live is available through our Overleaf Labs program. For most requirements, the standard 2025 TeX Live image should be sufficient, but for those who wish to use the most recent features provided by the $\text{\LaTeX}$ team, the rolling version provided by Overleaf Labs may be required.

1.2 Including DocumentMetadata

A key requirement for setting up the tagging support is to include the `DocumentMetadata` prior to the `\documentclass` declaration. Please see the $\text{\LaTeX}$ team’s tagging usage instructions for details.

```

\DocumentMetadata{
  lang          = en,
  pdfstandard   = ua-2,
  pdfstandard   = a-4,
  tagging=on,
  tagging-setup={math/setup=mathml-SE}
}

```

Figure 1: The DocumentMetada used in this document

2 Tagging special structures

The automatic tagging provided now in $\text{\LaTeX}$ doesn't guarantee accessibility. Having an error-free document, that uses standard $\text{\LaTeX}$ commands (for sections, lists, etc.) is also essential.

2.1 Remember to include alternate text for images

For figures, you can use the `alt` optional parameter to the `\includegraphics` command that is provided with the `graphicx` package.

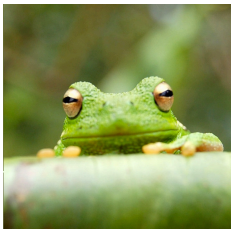


Figure 2: This image has some alt text

2.2 Tagging tables

Use the `table` and `tabular` environments for basic tables — see Table 1.

For tables, you should include the command that will identify the header row, which would typically be the first row: `\tagpdfsetup{table/header-rows={1}}`.

Item	Quantity
Widgets	42
Gadgets	13

Table 1: An example table.

2.3 How to write accessible mathematics

$\text{\LaTeX}$ is great at typesetting mathematics.

For equations, you should compile with `LuaLaTeX` if possible to generate the `MathML`. If you compile with `LuaLaTeX` and include the `tagging-setup={math/setup=mathml-SE}` option, `MathML` versions of the equations will be added into the tagged PDF directly.

Below is some example mathematics, which will be properly tagged and provided with its `MathML` translation in the final PDF

Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_i^n X_i$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

3 Other accessibility considerations

While PDF tagging is a key criteria that many accessibility verification systems check for, accessibility criteria and testing can vary, and the efficacy of tagged PDFs can vary with different assistive technologies.

It is recommended that you validate sample tagged PDFs according to the criteria set by your institution, and test using the assistive technologies that have been adopted by your users.

4 Other formats

In addition to PDF output, it is also possible to compile to HTML using `latexml` and `bookml`.

When using Overleaf, the GitHub synchronization feature makes it possible to run these tools outside of Overleaf as GitHub workflows.

References

- [LaT25] LaTeX team. The latex tagged pdf project, 2025. <https://latex3.github.io/tagging-project/>.
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